

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – III

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Date of Assignment: 14/07/2026	Date/Time of Submission: 14/07/2026

AIM: Write the program for the following.

A. Create methods add(). multiply(), subtract() ,divide() with suitable parameters and call these methods using concept of C# delegate.

ALGORITHM:

CODE:

```
using System;
delegate void Operation(int a, int b);
class Calculator
{
    public void Add(int a , int b)
    {
        Console.WriteLine("Addition: " + (a + b));
    }

    public void Sub(int a, int b)
    {
        Console.WriteLine("Subtraction: " + (a - b));
    }

    public void Mul(int a, int b)
    {
        Console.WriteLine("Multiplication: " + (a * b));
    }

    public void Div(int a, int b)
    {
        Console.WriteLine("Division: " + (a / b));
    }
}
class Practical_3A
{
    public static void Main(string[] args)
    {
        Calculator c = new Calculator();
        Operation obj;

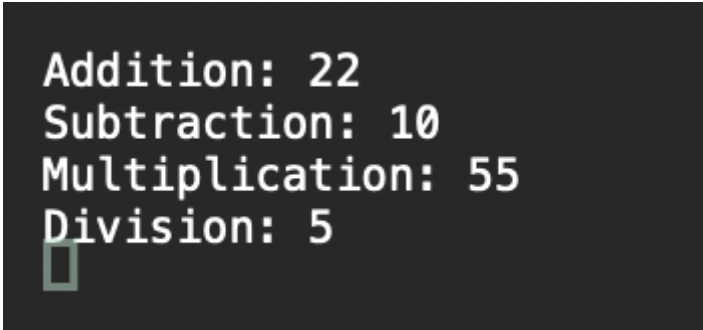
        obj = c.Add;
        obj(11, 11);

        obj = c.Sub;
        obj(20, 10);

        obj = c.Mul;
        obj(5, 11);
    }
}
```

```
        obj = c.Div;  
        obj(20, 4);  
        Console.ReadLine();  
    }  
}
```

OUTPUT:



```
Addition: 22  
Subtraction: 10  
Multiplication: 55  
Division: 5
```

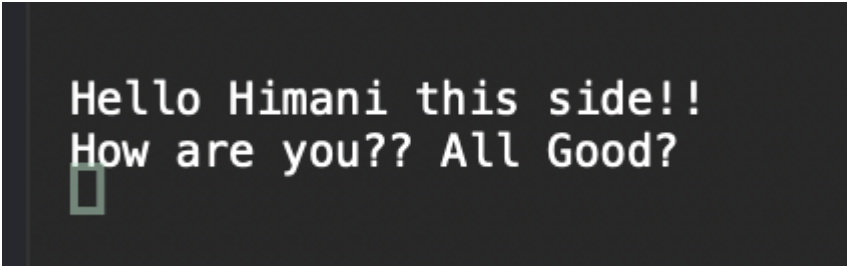
B. Write a program using multicast delegate.

ALGORITHM:

CODE:

```
using System;
delegate void Message();
class Program
{
    public void Hello()
    {
        Console.WriteLine("Hello Himani this side!!");
    }
    public void Fine()
    {
        Console.WriteLine("How are you?? All Good?");
    }
    static void Main(string[] args)
    {
        Program p = new Program();
        Message msg = p.Hello;
        msg += p.Fine;
        msg();
        Console.ReadLine();
    }
}
```

OUTPUT:

A screenshot of a console window with a dark background and light green text. It displays the output of the program: "Hello Himani this side!!" on the first line and "How are you?? All Good?" on the second line. A small green cursor is visible at the end of the second line.

```
Hello Himani this side!!
How are you?? All Good?
```

C. Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

ALGORITHM:

CODE:

```
using System;
class BankAccount
{
    public int AccountNumber { get; set; }
    private double balance;
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (value >= 0)
                balance = value;
            else
                Console.WriteLine("Balance cannot be negative.");
        }
    }
}
class Program
{
    static void Main(string[] args)
    {
        BankAccount b = new BankAccount();

        b.AccountNumber = 1001;
        b.Balance = 5000;

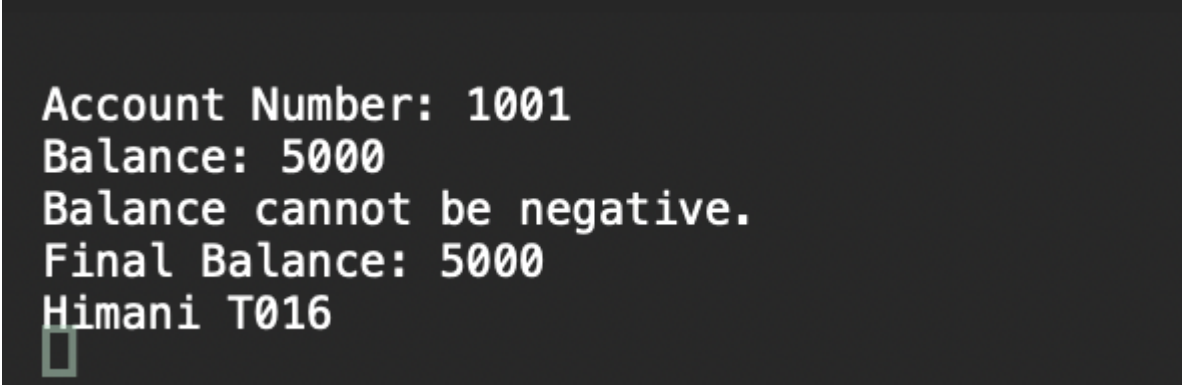
        Console.WriteLine("Account Number: " + b.AccountNumber);
        Console.WriteLine("Balance: " + b.Balance);

        b.Balance = -2000;

        Console.WriteLine("Final Balance: " + b.Balance);
        Console.WriteLine("Himani T016");

        Console.ReadLine();
    }
}
```

OUTPUT:

A screenshot of a terminal window with a dark background and light green text. The output shows the program's execution: 'Account Number: 1001', 'Balance: 5000', 'Balance cannot be negative.', 'Final Balance: 5000', and 'Himani T016'. A small green cursor is visible at the bottom left.

```
Account Number: 1001
Balance: 5000
Balance cannot be negative.
Final Balance: 5000
Himani T016
```